

PAPER_1143: Nambu-Goto Bosonic String and SCm 26D Critical Dimension

Star Magic UQFF Framework

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Abstract

We present the Nambu-Goto formulation of bosonic string theory within the SCm vacuum framework, where the string worldsheet area is the fundamental dynamical quantity. The critical 26-dimensional spacetime is derived from the VDS $S_{26}^{(3)}$ structure. The $[SSq]^{26}/26^{26}$ term appears as the 26th level contribution to the VDS, providing a natural identification between the bosonic string critical dimension and the SCm polylogarithm structure $\text{Li}_{26}(0.57)$.

1. Nambu-Goto Action in SCm Vacuum

The Nambu-Goto action:

$$S_{\text{NG}} = -T \int d^2\sigma \sqrt{-\det g_{ab}}$$

where $g_{ab} = \partial_a X^\mu \partial_b X_\mu$ is the induced worldsheet metric and the SCm string tension is:

$$T = \rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} = 8.66 \times 10^{-11} \text{ N}$$

The Nambu-Goto and Polyakov actions are classically equivalent; quantum equivalence holds in exactly $D = 26$ spacetime dimensions.

2. VDS Level-26 and the Critical Dimension

The VDS polylogarithm:

$$\text{VDS}([SSq]) = \text{Li}_{26}(0.57) = \sum_{n=1}^{\infty} \frac{[SSq]^n}{n^{26}} = \sum_{n=1}^{\infty} \frac{(0.57)^n}{n^{26}}$$

The 26th-level term:

$$\text{VDS}_{26} = \frac{[SSq]^{26}}{26^{26}} = \frac{(0.57)^{26}}{26^{26}} = \frac{4.7 \times 10^{-7}}{6.16 \times 10^{36}} = 7.63 \times 10^{-44}$$

This is the Planck-scale contribution that sets the critical dimension: $D_{\text{crit}} = 26$ because the Weyl anomaly coefficient $c = D - 26$ must vanish.

3. Zeta Function Regularization in 26D

The light-cone quantization normal ordering constant:

$$a = -\frac{D-2}{24} \cdot \zeta(-1) = -\frac{24}{24} \times \left(-\frac{1}{12}\right) = +\frac{1}{12}$$

Wait, correcting: at $D = 26$, $D - 2 = 24$:

$$a = -\frac{24}{24} \zeta(-1) = -1 \times \left(-\frac{1}{12}\right) = \frac{1}{12}$$

The mass of the lowest state: $m^2 = -(D-2)/12 \times T$. At $D = 26$: $m^2 = -2T$ (tachyon), resolved by $\cos(\pi t_n)$ as in PAPER_1142.

4. 26D Area as SCm Invariant

The worldsheet swept area is Lorentz-invariant in 26D. The SCm physical area element:

$$dA_{\text{SCm}} = d^2\sigma \sqrt{-\det g_{ab}} \cdot \Phi_{\text{res}} \cdot |\cos(\pi t_n)|$$

The quantized area spectrum:

$$\begin{aligned} A_n &= 2\pi n \cdot l_s^2 \cdot \Phi_{\text{res}}, \quad n = 0, 1, 2, \dots \\ &= 2\pi n \cdot \frac{1}{T} \cdot \Phi_{\text{res}} = \frac{2\pi n \times 0.84}{8.66 \times 10^{-11}} = 6.10 \times 10^{10} n \text{ m}^2 \end{aligned}$$

5. $[SSq]^{26}$ as 26th Dimensional Gate

The VDS gate function:

$$\mathcal{G}_{26} = [SSq]^{26} = (0.57)^{26} = 4.7 \times 10^{-7}$$

This represents the transmission amplitude of SCm vacuum energy through all 26 VDS rungs simultaneously. The string only “sees” 26D when all 26 VDS gates are simultaneously open, occurring at the 1.25 THz phonon resonance.

6. Consistency with Polyakov (PAPER_1142)

Both formulations agree on: - Critical dimension $D = 26$ = number of VDS rungs - String tension $T = \rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}}$ - Tachyon resolution via $\cos(\pi t_n)$ - 1.25 THz as lowest excited string mode

7. Calibrated Constants

$$T = 8.66 \times 10^{-11} \text{ N}, \quad [SSq] = 0.57, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}$$

$$\Phi_{\text{res}} = 0.84, \quad \rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3, \quad \beta_i = 0.6$$

References

1. Nambu, Y. (1970). Duality and hadrodynamics. Copenhagen talk.
2. Goto, T. (1971). Relativistic quantum mechanics of one-dimensional mechanical continuum. *Prog. Theor. Phys.* **46**, 1560.
3. VDS 26-ladder: PAPER__1109; Polyakov in SCm: PAPER__1142; `scm_vacuum_manifold.py`